Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Van Steyn Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
13039 Pellandini RD	Galt	Sacramento	95632-8818

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1958

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X146-X230-X014-XXXX X146-X230-X015-XXXX

B. OPERATORS

Van Steyn, Case A

Operator name: Van Steyn, Case A

Telephone no.: (209) 745-1428

Mailing Address Number and Street	City	State	Zip Code
13039 Pellandini RD	Galt	CA	95632

Landline	Cellular
(209) 745-1428	

This operator is responsible for paying permit fees.

C. OWNERS

Van Steyn, Case A

Legal owner name: Van Steyn, Case A

Telephone no.: (209) 745-1428

Mailing Address Number and Street	City	State	Zip Code
13039 Pellandini RD	Galt	CA	95632

Landline	Cellular
(209) 745-1428	

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	525	50	0	0	0	0
Maximum number	525	50	0	0	0	0
Average number	525	50	0	0	0	0
Avg live weight (lbs)	1,350	1,450	0	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 14,073.58 *tons per reporting period*

Total nitrogen from manure: 183,588.37 *lbs per reporting period*

After ammonia losses (30% loss applied): 128,511.86 *lbs per reporting period*

Total phosphorus from manure: 30,716.61 *lbs per reporting period*

Total potassium from manure: 99,852.49 *lbs per reporting period*

Total salt from manure: 258,693.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,395,000 *gallons*

Total nitrogen generated: 29,564.58 *lbs*

Total phosphorus generated: 2,933.86 *lbs*

Total potassium generated: 33,994.90 *lbs*

Total salt generated: 330,852.55 *lbs*

	7,395,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,395,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy/Dom.	Ground water
G30 Irr. Well	Ground water
G8 Irr. Well	Ground water
North Irr. Well	Ground water
South Irr. Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/15/2023	Solid commercial fertilizer *Starter Fertilizer*	1.10 *ton*	Dry-weight	8.0	16.000000	38.000000	38.000000	0.000000
08/01/2023	Liquid commercial fertilizer *Commercial Fertilizer Side dress*	830.00 *gal*			90.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	6,557.56	769.12	769.12	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	6,557.56	769.12	769.12	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	Corral solids	775.00 *ton*	Dry-weight	79.1		26,100.00	4,730.00	5,340.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	8,455.10	1,532.28	1,729.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	8,455.10	1,532.28	1,729.89	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
G01	11	11	2	process wastewater	X146-X230-X014-XXXX
G02	17	17	2	process wastewater	X146-X230-X006-XXXX
G03	23	23	2	process wastewater	X146-X230-X014-XXXX
G05	27	27	1	process wastewater	X146-X230-X014-XXXX
G06	11	11	2	process wastewater	X146-X230-X014-XXXX
G08	12	12	0	none	X146-X230-X015-XXXX
G09	3	3	1	manure	X146-X230-X014-XXXX X146-X230-X015-XXXX
G11	19	19	2	manure	X146-X230-X009-XXXX X146-X230-X015-XXXX
G12	8	8	2	manure	X146-X230-X015-XXXX
G13	26	26	2	manure	X146-X230-X009-XXXX
G14	15	15	2	manure	X146-X230-X009-XXXX
G30	30	30	0	none	X148-X061-X022-XXXX
Totals for areas that were used for application	160	160	18		
Totals for areas that were not used for application	42	42	0		
Land application area totals	202	202	18		

B. CROPS AND HARVESTS

G01

Field name: G01

10/29/2022: Winter Forage

Crop: Winter Forage Acres planted: 11 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	159.50 *ton*	Dry-weight		51.4	22,600.00	2,580.00	7,860.00		7.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	308.00	36.40	106.40	0.00
Total actual harvest content	14.50	318.52	36.36	110.78	1,076.78

G01

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	313.50 *ton*	Dry-weight		69.6	11,000.00	2,140.00	10,700.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	44.80	179.20	0.00
Total actual harvest content	28.50	190.61	37.08	185.41	1,091.66

G02

Field name: G02

10/28/2022: Ryegrass, silage

Crop: Ryegrass, silage Acres planted: 17 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	238.00 *ton*	Dry-weight		82.0	21,500.00	3,680.00	28,400.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	107.80	19.60	142.80	0.00
Total actual harvest content	14.00	108.36	18.55	143.14	574.56

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	484.50 *ton*	Dry-weight		69.6	11,000.00	2,140.00	10,700.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	179.20	0.00
Total actual harvest content	28.50	190.61	37.08	185.41	1,091.66

G03

Field name: G03

10/28/2022: Ryegrass, silage

Crop: Ryegrass, silage Acres planted: 23 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	322.00 *ton*	Dry-weight		82.0	21,500.00	3,680.00	28,400.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	107.80	19.60	142.80	0.00
Total actual harvest content	14.00	108.36	18.55	143.14	574.56

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	655.50 *ton*	Dry-weight		71.6	10,800.00	2,350.00	12,800.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	44.80	201.60	0.00
Total actual harvest content	28.50	174.83	38.04	207.21	1,003.66

G05

Field name: G05

10/27/2022: Ryegrass, silage

Crop: Ryegrass, silage Acres planted: 27 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	378.00 *ton*	Dry-weight		75.4	26,800.00	5,200.00	44,000.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	184.80	35.00	196.00	0.00
Total actual harvest content	14.00	184.60	35.82	303.07	902.33

G06

Field name: G06

10/27/2022: Ryegrass, silage

Crop: Ryegrass, silage Acres planted: 11 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	154.00 *ton*	Dry-weight		75.4	26,800.00	5,200.00	44,000.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	184.80	35.00	196.00	0.00
Total actual harvest content	14.00	184.60	35.82	303.07	902.33

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	313.50 *ton*	Dry-weight		71.6	10,800.00	2,350.00	12,800.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	44.80	201.60	0.00
Total actual harvest content	28.50	174.83	38.04	207.21	1,003.66

G09

Field name: G09

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 3 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	85.50 *ton*	Dry-weight		69.2	10,900.00	1,950.00	5,420.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	191.36	34.23	95.15	1,070.92

G11

Field name: G11

10/29/2022: Winter Forage

Crop: Winter Forage Acres planted: 19 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	275.50 *ton*	Dry-weight		51.4	22,600.00	2,580.00	7,860.00		7.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	308.00	36.40	106.40	0.00
Total actual harvest content	14.50	318.52	36.36	110.78	1,076.78

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	541.50 *ton*	Dry-weight		69.2	10,900.00	1,950.00	5,420.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	191.36	34.23	95.15	1,070.92

G12

Field name: G12

10/30/2022: Winter Forage

Crop: Winter Forage Acres planted: 8 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	116.00 *ton*	Dry-weight		51.4	22,600.00	2,580.00	7,860.00		7.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	308.00	36.40	106.40	0.00
Total actual harvest content	14.50	318.52	36.36	110.78	1,076.78

G12

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	228.00 *ton*	Dry-weight		69.2	10,900.00	1,950.00	5,420.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.50	191.36	34.23	95.15	1,070.92

G13

Field name: G13

10/31/2022: Oats, hay

Crop: Oats, hay Acres planted: 26 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	104.00 *ton*	Dry-weight		12.0	19,900.00	1,250.00	7,850.00		6.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	140.10	8.80	55.26	454.08

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	741.00 *ton*	Dry-weight		67.2	9,720.00	1,250.00	4,800.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	39.20	182.00	0.00
Total actual harvest content	28.50	181.73	23.37	89.74	1,159.15

G14

Field name: G14

10/31/2022: Oats, hay

Crop: Oats, hay Acres planted: 15 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	60.00 *ton*	Dry-weight		12.0	19,900.00	1,250.00	7,850.00		6.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	140.10	8.80	55.26	454.08

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	427.50 *ton*	Dry-weight		67.2	9,720.00	1,250.00	4,800.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	39.20	182.00	0.00
Total actual harvest content	28.50	181.73	23.37	89.74	1,159.15

NUTRIENT BUDGET

A. LAND APPLICATIONS

G01 - 10/29/2022: Winter Forage

Field name: G01

Crop: Winter Forage — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.30	0.00	0.00	0.00	1,218,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	69.14	6.58	86.51	713.73	210,000.00 *gal*
Application event totals		69.14	6.58	86.51	713.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Pipeline	Steady rain	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	165,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Pipeline	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	165,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

G01 - 06/12/2023: Corn, silage

Field name: G01

Crop: Corn, silage — Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

G01 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.47	0.00	0.00	0.00	1,932,000.00 *gal*
Application event totals		0.47	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	82.05	5.64	88.10	774.26	210,000.00 *gal*
North Irr. Well	Ground water	0.40	0.00	0.00	0.00	1,638,000.00 *gal*
Application event totals		82.44	5.64	88.10	774.26	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	82.05	5.64	88.10	774.26	210,000.00 *gal*
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		82.33	5.64	88.10	774.26	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	82.05	5.64	88.10	774.26	210,000.00 *gal*
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		82.33	5.64	88.10	774.26	

G01 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

G02 - 10/28/2022: Ryegrass, silage

Field name: G02

Crop: Ryegrass, silage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	47.93	4.56	59.97	494.81	225,000.00 *gal*
Application event totals		47.93	4.56	59.97	494.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Pipeline	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	40.29	6.17	48.86	592.00	180,000.00 *gal*
Application event totals		40.29	6.17	48.86	592.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Pipeline	No precipitation	No precipitation	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	53.72	8.22	65.15	789.34	240,000.00 *gal*
Application event totals		53.72	8.22	65.15	789.34	

G02 - 06/16/2023: Corn, silage

Field name: G02

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	91.01	6.26	97.72	858.85	360,000.00 *gal*
North Irr. Well	Ground water	0.40	0.00	0.00	0.00	2,520,000.00 *gal*
Application event totals		91.41	6.26	97.72	858.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.50	3.13	48.86	429.42	180,000.00 *gal*
North Irr. Well	Ground water	0.32	0.00	0.00	0.00	2,058,000.00 *gal*
Application event totals		45.83	3.13	48.86	429.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	75.84	5.21	81.44	715.71	300,000.00 *gal*
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		76.12	5.21	81.44	715.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.50	3.13	48.86	429.42	180,000.00 *gal*
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		45.78	3.13	48.86	429.42	

G02 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

G03 - 10/28/2022: Ryegrass, silage

Field name: G03

Crop: Ryegrass, silage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	42.52	4.05	53.19	438.87	270,000.00 *gal*
North Irr. Well	Ground water	0.27	0.00	0.00	0.00	2,310,000.00 *gal*
Application event totals		42.78	4.05	53.19	438.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	345,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Pipeline	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	39.71	6.08	48.15	583.42	240,000.00 *gal*
Application event totals		39.71	6.08	48.15	583.42	

G03 - 06/19/2023: Corn, silage

Field name: G03

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	103.70	7.13	111.36	978.65	555,000.00 *gal*
North Irr. Well	Ground water	0.37	0.00	0.00	0.00	3,192,000.00 *gal*
Application event totals		104.08	7.13	111.36	978.65	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.32	0.00	0.00	0.00	2,772,000.00 *gal*
Application event totals		0.32	0.00	0.00	0.00	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	2,394,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	78.48	5.39	84.27	740.60	420,000.00 *gal*
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	2,394,000.00 *gal*
Application event totals		78.76	5.39	84.27	740.60	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	2,394,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

G03 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	50.45	3.47	54.17	476.10	270,000.00 *gal*
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	2,394,000.00 *gal*
Application event totals		50.73	3.47	54.17	476.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.28	0.00	0.00	0.00	2,394,000.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

G05 - 10/27/2022: Ryegrass, silage

Field name: G05

Crop: Ryegrass, silage

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	88.53	8.42	110.77	913.87	660,000.00 *gal*
North Irr. Well	Ground water	0.27	0.00	0.00	0.00	2,772,000.00 *gal*
Application event totals		88.81	8.42	110.77	913.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	405,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

G05 - 10/27/2022: Ryegrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Pipeline	Steady rain	No precipitation	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	38.05	5.82	46.15	559.12	270,000.00 *gal*
Application event totals		38.05	5.82	46.15	559.12	

G06 - 10/27/2022: Ryegrass, silage

Field name: G06

Crop: Ryegrass, silage

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	69.14	6.58	86.51	713.73	210,000.00 *gal*
Application event totals		69.14	6.58	86.51	713.73	

02/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	165,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

03/09/2023	Pipeline	No precipitation	Steady rain	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	57.08	8.74	69.22	838.67	165,000.00 *gal*
Application event totals		57.08	8.74	69.22	838.67	

G06 - 06/24/2023: Corn, silage

Field name: G06

Crop: Corn, silage

Plant date: 06/24/2023

G06 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	117.21	8.06	125.86	1,106.09	300,000.00 *gal*
North Irr. Well	Ground water	0.47	0.00	0.00	0.00	1,932,000.00 *gal*
Application event totals		117.68	8.06	125.86	1,106.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.40	0.00	0.00	0.00	1,638,000.00 *gal*
Application event totals		0.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	70.33	4.83	75.51	663.66	180,000.00 *gal*
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		70.61	4.83	75.51	663.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	41.02	2.82	44.05	387.13	105,000.00 *gal*
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		41.31	2.82	44.05	387.13	

G06 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
North Irr. Well	Ground water	0.29	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

G09 - 06/19/2023: Corn, silage

Field name: G09

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	130.92	23.73	26.79	0.00	36.00 *ton*
Application event totals		130.92	23.73	26.79	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	223.44	535,500.00 *gal*
Application event totals		0.00	0.00	0.00	223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

G09 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	202.16	484,500.00 *gal*
Application event totals		0.00	0.00	0.00	202.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		0.00	0.00	0.00	159.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SIDE DRESS	Liquid commercial fertilizer	75.00	0.00	0.00	0.00	
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		75.00	0.00	0.00	159.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		0.00	0.00	0.00	159.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		0.00	0.00	0.00	159.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		0.00	0.00	0.00	159.60	

G09 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	159.60	382,500.00 *gal*
Application event totals		0.00	0.00	0.00	159.60	

G11 - 10/29/2022: Winter Forage

Field name: G11

Crop: Winter Forage Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	134.04	21.97	17.67	1,252.50	250.00 *ton*
Application event totals		134.04	21.97	17.67	1,252.50	

G11 - 06/20/2023: Corn, silage

Field name: G11

Crop: Corn, silage Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	143.55	26.02	29.37	0.00	250.00 *ton*
Application event totals		143.55	26.02	29.37	0.00	

G11 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	184.80	2,805,000.00 *gal*
Application event totals		0.00	0.00	0.00	184.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	161.28	2,448,000.00 *gal*
Application event totals		0.00	0.00	0.00	161.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.72	2,014,500.00 *gal*
Application event totals		0.00	0.00	0.00	132.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.69	2,014,000.00 *gal*
Application event totals		0.00	0.00	0.00	132.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SIDE DRESS	Liquid commercial fertilizer	75.00	0.00	0.00	0.00	
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.72	2,014,500.00 *gal*
Application event totals		75.00	0.00	0.00	132.72	

G11 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.72	2,014,500.00 *gal*
Application event totals		0.00	0.00	0.00	132.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.72	2,014,500.00 *gal*
Application event totals		0.00	0.00	0.00	132.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	132.72	2,014,500.00 *gal*
Application event totals		0.00	0.00	0.00	132.72	

G12 - 10/30/2022: Winter Forage

Field name: G12

Crop: Winter Forage Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	127.34	20.88	16.78	1,189.88	100.00 *ton*
Application event totals		127.34	20.88	16.78	1,189.88	

G12 - 06/18/2023: Corn, silage

Field name: G12

Crop: Corn, silage Plant date: 06/18/2023

G12 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	136.37	24.71	27.90	0.00	100.00 *ton*
Application event totals		136.37	24.71	27.90	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	207.48	1,326,000.00 *gal*
Application event totals		0.00	0.00	0.00	207.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	175.56	1,122,000.00 *gal*
Application event totals		0.00	0.00	0.00	175.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.64	

G12 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SIDE DRESS	Liquid commercial fertilizer	75.00	0.00	0.00	0.00	
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		75.00	0.00	0.00	143.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
G8 Irr. Well	Ground water	0.00	0.00	0.00	143.64	918,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.64	

G13 - 10/31/2022: Oats, hay

Field name: G13

Crop: Oats, hay

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	137.13	22.48	18.07	1,281.40	350.00 *ton*
Application event totals		137.13	22.48	18.07	1,281.40	

G13 - 06/17/2023: Corn, silage

Field name: G13

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	117.49	21.29	24.04	0.00	280.00 *ton*
Application event totals		117.49	21.29	24.04	0.00	

06/17/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	7.76	0.00	0.00	0.00	3,442,500.00 *gal*
Application event totals		7.76	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.89	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		6.89	0.00	0.00	0.00	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.15	0.00	0.00	0.00	2,728,500.00 *gal*
Application event totals		6.15	0.00	0.00	0.00	

G13 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.15	0.00	0.00	0.00	2,728,500.00 *gal*
Application event totals		6.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SIDE DRESS	Liquid commercial fertilizer	75.00	0.00	0.00	0.00	
South Irr. Well	Ground water	6.15	0.00	0.00	0.00	2,728,500.00 *gal*
Application event totals		81.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.15	0.00	0.00	0.00	2,728,500.00 *gal*
Application event totals		6.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.15	0.00	0.00	0.00	2,728,500.00 *gal*
Application event totals		6.15	0.00	0.00	0.00	

G14 - 10/31/2022: Oats, hay

Field name: G14

Crop: Oats, hay

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	135.83	22.27	17.90	1,269.20	200.00 *ton*
Application event totals		135.83	22.27	17.90	1,269.20	

G14 - 06/11/2023: Corn, silage

Field name: G14

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	109.10	19.77	22.32	0.00	150.00 *ton*
Application event totals		109.10	19.77	22.32	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	9.16	0.00	0.00	0.00	2,346,000.00 *gal*
Application event totals		9.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	8.17	0.00	0.00	0.00	2,091,000.00 *gal*
Application event totals		8.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		6.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		6.37	0.00	0.00	0.00	

G14 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
SIDE DRESS	Liquid commercial fertilizer	75.00	0.00	0.00	0.00	
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		81.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		6.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		6.37	0.00	0.00	0.00	

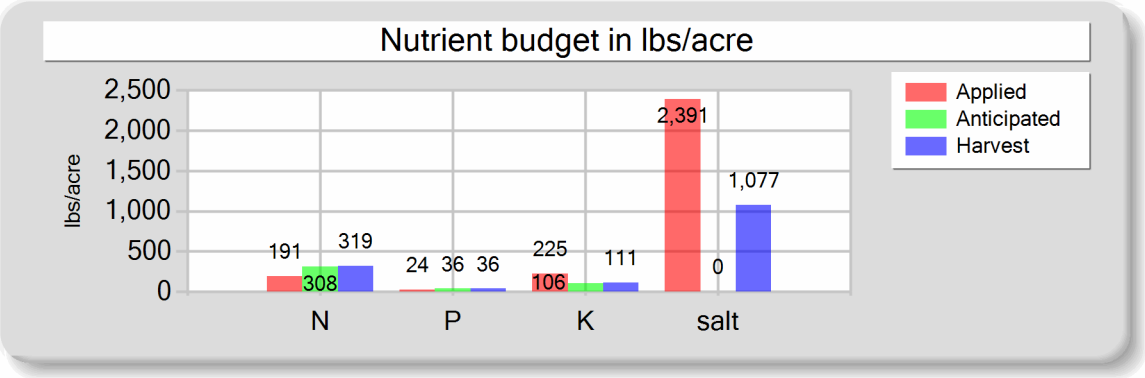
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
South Irr. Well	Ground water	6.37	0.00	0.00	0.00	1,632,000.00 *gal*
Application event totals		6.37	0.00	0.00	0.00	

B. NUTRIENT BUDGET

G01 - 10/29/2022: Winter Forage

Field name: G01 Crop: Winter Forage Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.30	24.05	224.95	2,391.07
Fresh water	0.30	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.60	24.05	224.95	2,391.07
Anticipated crop nutrient removal	308.00	36.40	106.40	0.00
Actual crop nutrient removal	318.52	36.36	110.78	1,076.78
Nutrient balance	-127.93	-12.31	114.17	1,314.29
Applied to removed ratio	0.60	0.66	2.03	2.22

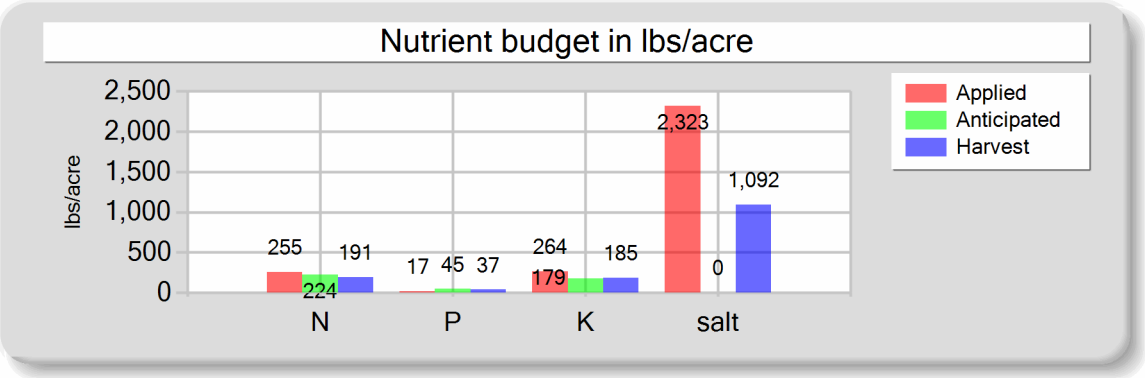
Fresh water applied	
1,218,000.00	*gallons*
44.85	*acre-inches*
4.08	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

G01 - 06/12/2023: Corn, silage

Field name: G01 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.14	16.92	264.30	2,322.79
Fresh water	2.29	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.43	16.92	264.30	2,322.79
Anticipated crop nutrient removal	224.00	44.80	179.20	0.00
Actual crop nutrient removal	190.61	37.08	185.41	1,091.66
Nutrient balance	64.83	-20.16	78.89	1,231.13
Applied to removed ratio	1.34	0.46	1.43	2.13

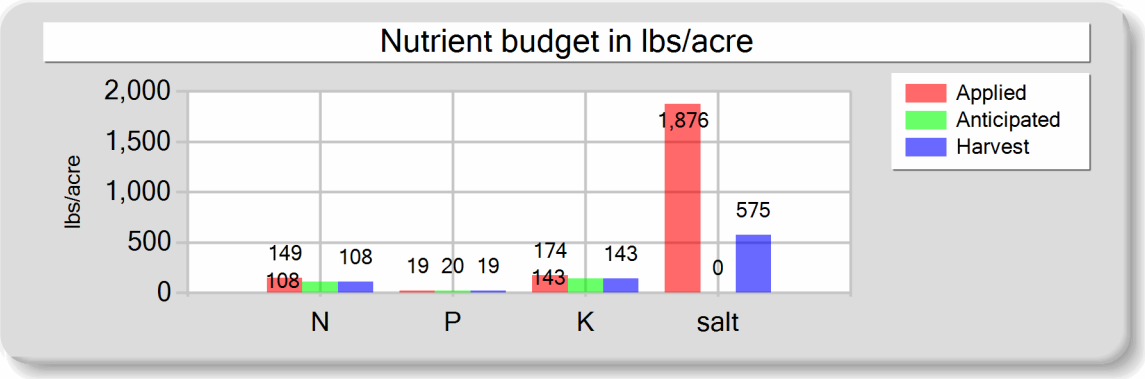
Fresh water applied	
9,450,000.00	*gallons*
348.01	*acre-inches*
31.64	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
2.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

G02 - 10/28/2022: Ryegrass, silage

Field name: G02 Crop: Ryegrass, silage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.95	18.95	173.99	1,876.15
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	148.95	18.95	173.99	1,876.15
Anticipated crop nutrient removal	107.80	19.60	142.80	0.00
Actual crop nutrient removal	108.36	18.55	143.14	574.56
Nutrient balance	40.59	0.41	30.85	1,301.59
Applied to removed ratio	1.37	1.02	1.22	3.27

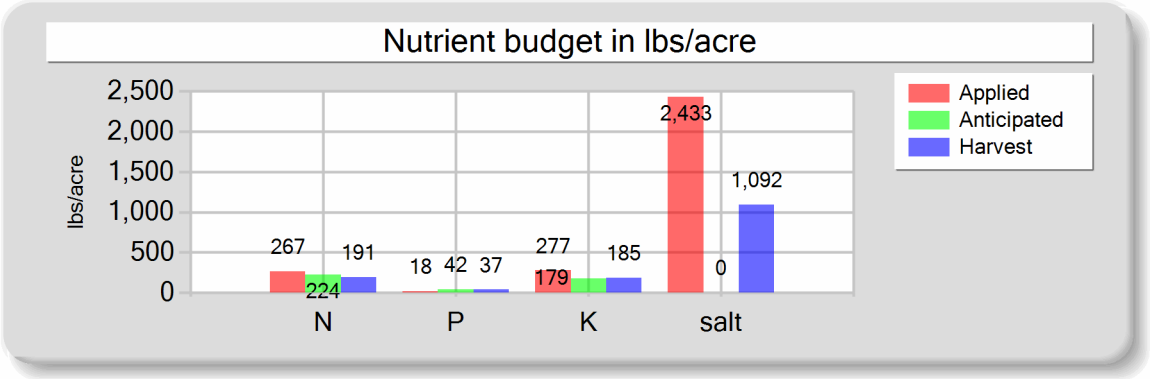
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
645,000.00	*gallons*
23.75	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

G02 - 06/16/2023: Corn, silage

Field name: G02 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	257.86	17.72	276.89	2,433.40
Fresh water	2.10	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.97	17.72	276.89	2,433.40
Anticipated crop nutrient removal	224.00	42.00	179.20	0.00
Actual crop nutrient removal	190.61	37.08	185.41	1,091.66
Nutrient balance	76.36	-19.36	91.48	1,341.74
Applied to removed ratio	1.40	0.48	1.49	2.23

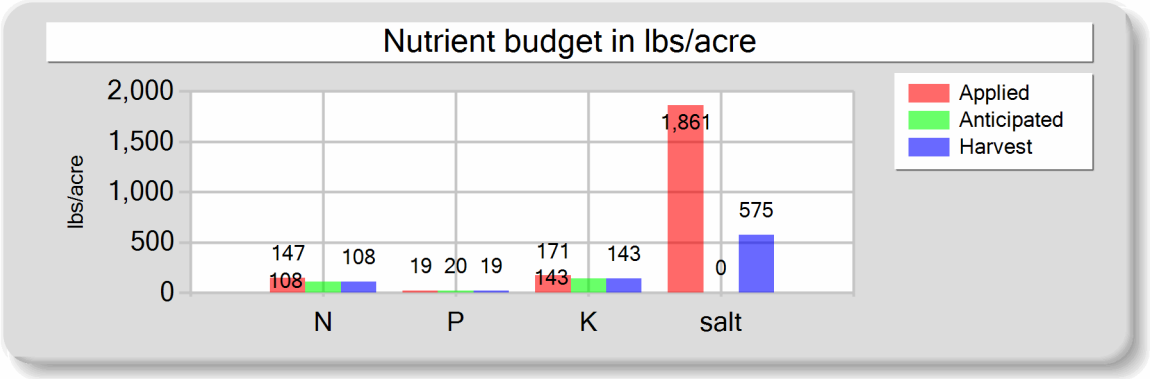
Fresh water applied	
13,398,000.00	*gallons*
493.40	*acre-inches*
29.02	*inches/acre*

Process wastewater applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

G03 - 10/28/2022: Ryegrass, silage

Field name: G03 Crop: Ryegrass, silage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	139.30	18.86	170.57	1,860.97
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.57	18.86	170.57	1,860.97
Anticipated crop nutrient removal	107.80	19.60	142.80	0.00
Actual crop nutrient removal	108.36	18.55	143.14	574.56
Nutrient balance	38.21	0.31	27.43	1,286.41
Applied to removed ratio	1.35	1.02	1.19	3.24

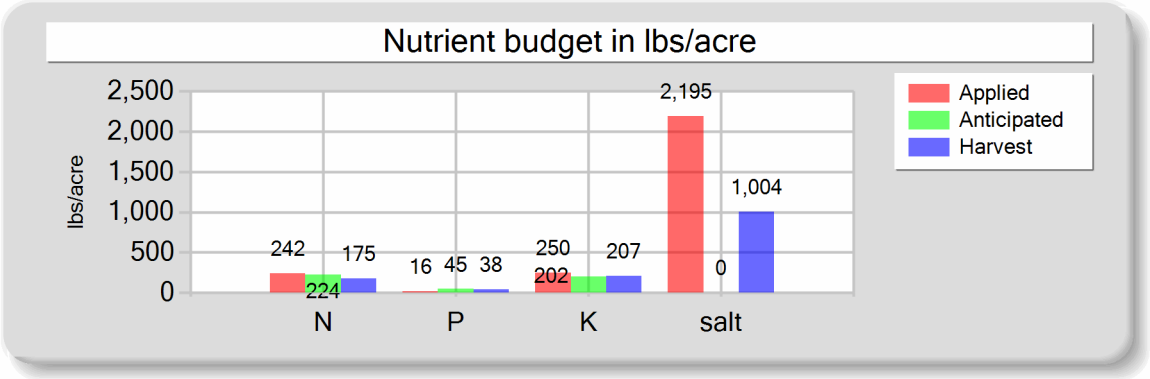
Fresh water applied	
2,310,000.00	*gallons*
85.07	*acre-inches*
3.70	*inches/acre*

Process wastewater applied	
855,000.00	*gallons*
31.49	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

G03 - 06/19/2023: Corn, silage

Field name: G03 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.64	15.99	249.80	2,195.35
Fresh water	2.08	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.72	15.99	249.80	2,195.35
Anticipated crop nutrient removal	224.00	44.80	201.60	0.00
Actual crop nutrient removal	174.83	38.04	207.21	1,003.66
Nutrient balance	66.89	-22.05	42.59	1,191.70
Applied to removed ratio	1.38	0.42	1.21	2.19

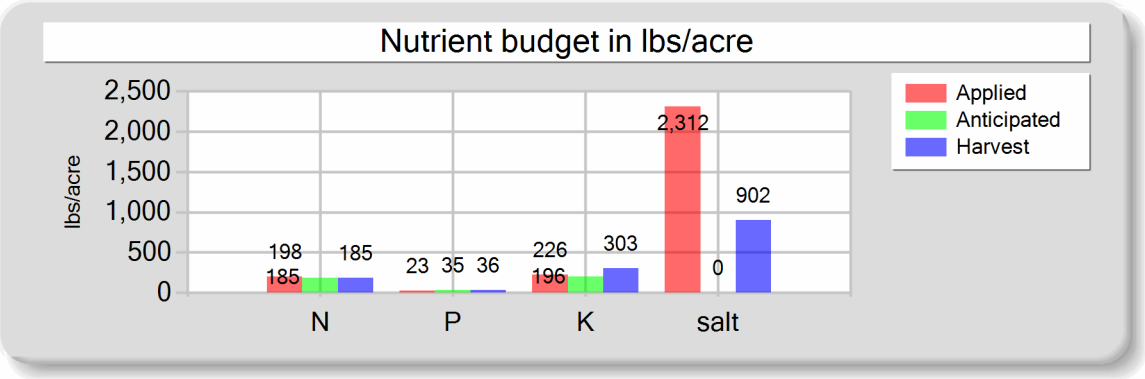
Fresh water applied	
17,934,000.00	*gallons*
660.45	*acre-inches*
28.72	*inches/acre*

Process wastewater applied	
1,245,000.00	*gallons*
45.85	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

G05 - 10/27/2022: Ryegrass, silage

Field name: G05 Crop: Ryegrass, silage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.66	22.99	226.14	2,311.66
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	197.94	22.99	226.14	2,311.66
Anticipated crop nutrient removal	184.80	35.00	196.00	0.00
Actual crop nutrient removal	184.60	35.82	303.07	902.33
Nutrient balance	13.34	-12.83	-76.94	1,409.33
Applied to removed ratio	1.07	0.64	0.75	2.56

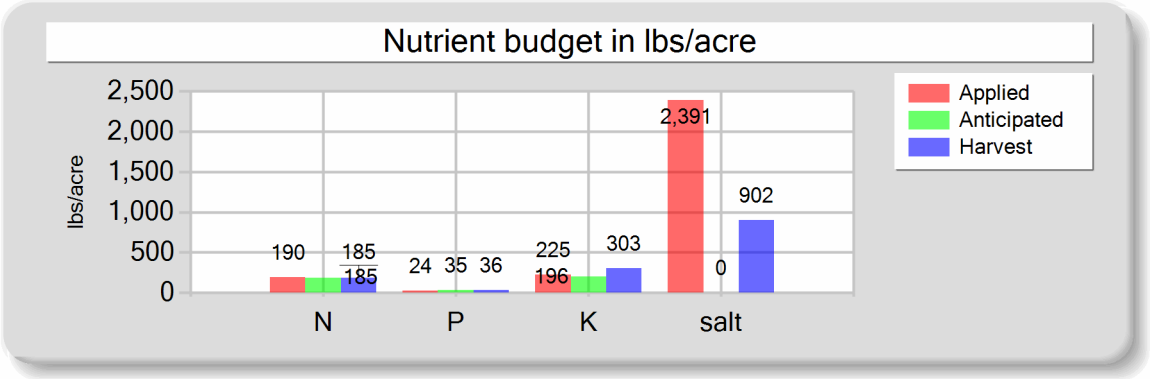
Fresh water applied	
2,772,000.00	*gallons*
102.08	*acre-inches*
3.78	*inches/acre*

Process wastewater applied	
1,335,000.00	*gallons*
49.16	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

G06 - 10/27/2022: Ryegrass, silage

Field name: G06 Crop: Ryegrass, silage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.30	24.05	224.95	2,391.07
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.30	24.05	224.95	2,391.07
Anticipated crop nutrient removal	184.80	35.00	196.00	0.00
Actual crop nutrient removal	184.60	35.82	303.07	902.33
Nutrient balance	5.70	-11.76	-78.12	1,488.74
Applied to removed ratio	1.03	0.67	0.74	2.65

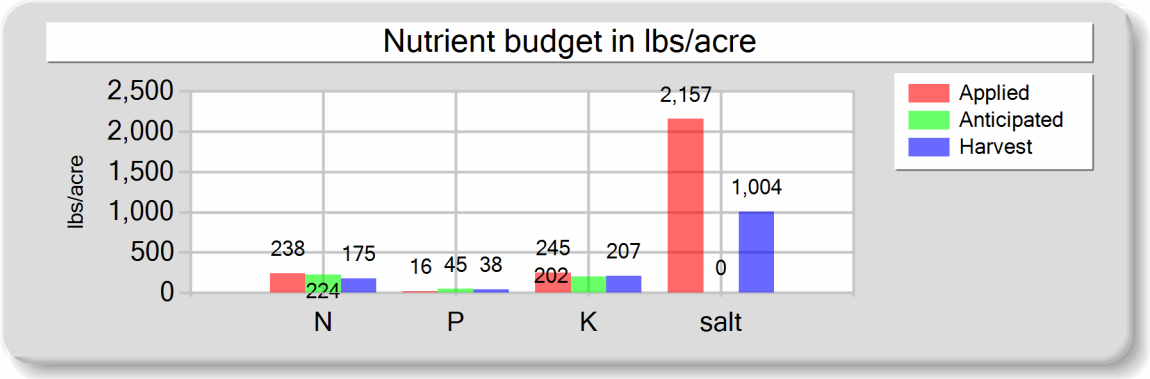
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

G06 - 06/24/2023: Corn, silage

Field name: G06 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.56	15.71	245.42	2,156.88
Fresh water	2.29	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.85	15.71	245.42	2,156.88
Anticipated crop nutrient removal	224.00	44.80	201.60	0.00
Actual crop nutrient removal	174.83	38.04	207.21	1,003.66
Nutrient balance	63.02	-22.33	38.22	1,153.22
Applied to removed ratio	1.36	0.41	1.18	2.15

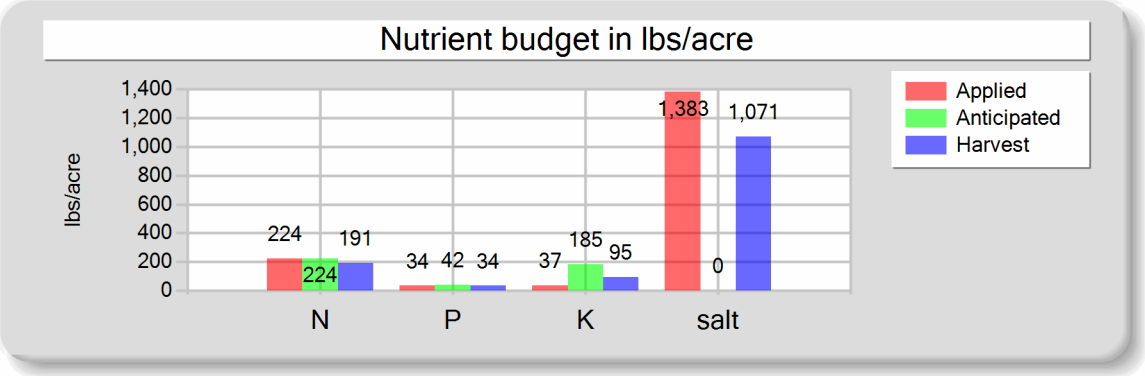
Fresh water applied	
9,450,000.00	*gallons*
348.01	*acre-inches*
31.64	*inches/acre*

Process wastewater applied	
585,000.00	*gallons*
21.54	*acre-inches*
1.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

G09 - 06/19/2023: Corn, silage

Field name: G09 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	79.00	10.00	10.00	0.00
Dry manure	130.92	23.73	26.79	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,383.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	223.92	33.73	36.79	1,383.18
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	191.36	34.23	95.15	1,070.92
Nutrient balance	32.56	-0.51	-58.37	312.27
Applied to removed ratio	1.17	0.99	0.39	1.29

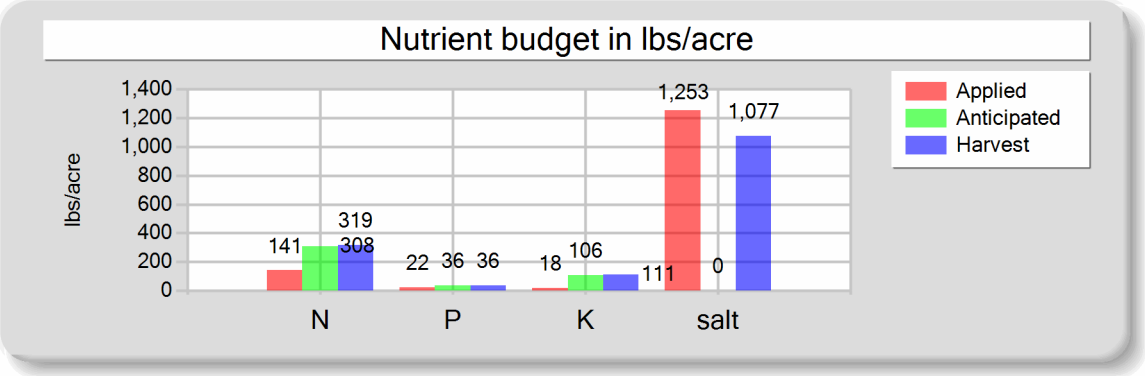
Fresh water applied
3,315,000.00 *gallons*
122.08 *acre-inches*
40.69 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G11 - 10/29/2022: Winter Forage

Field name: G11 Crop: Winter Forage Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	134.04	21.97	17.67	1,252.50
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.04	21.97	17.67	1,252.50
Anticipated crop nutrient removal	308.00	36.40	106.40	0.00
Actual crop nutrient removal	318.52	36.36	110.78	1,076.78
Nutrient balance	-177.48	-14.39	-93.11	175.72
Applied to removed ratio	0.44	0.60	0.16	1.16

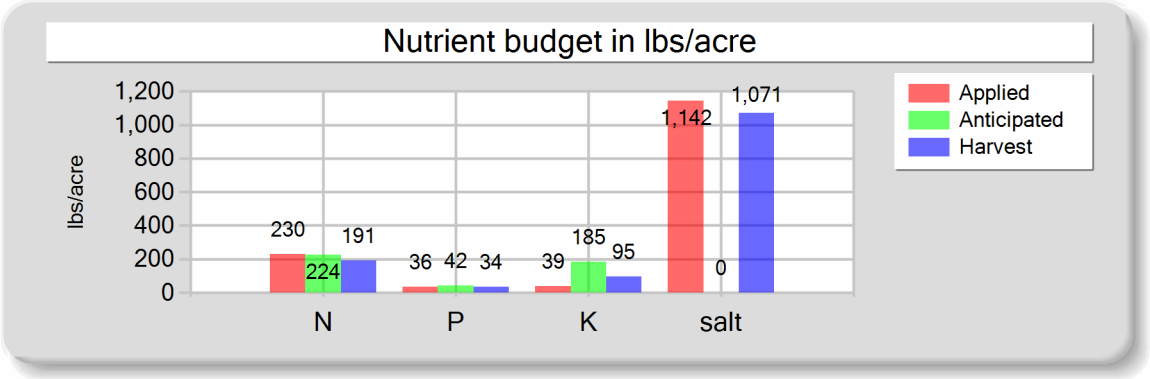
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G11 - 06/20/2023: Corn, silage

Field name: G11 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	79.00	10.00	10.00	0.00
Dry manure	143.55	26.02	29.37	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,142.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.55	36.02	39.37	1,142.35
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	191.36	34.23	95.15	1,070.92
Nutrient balance	38.19	1.78	-55.78	71.44
Applied to removed ratio	1.20	1.05	0.41	1.07

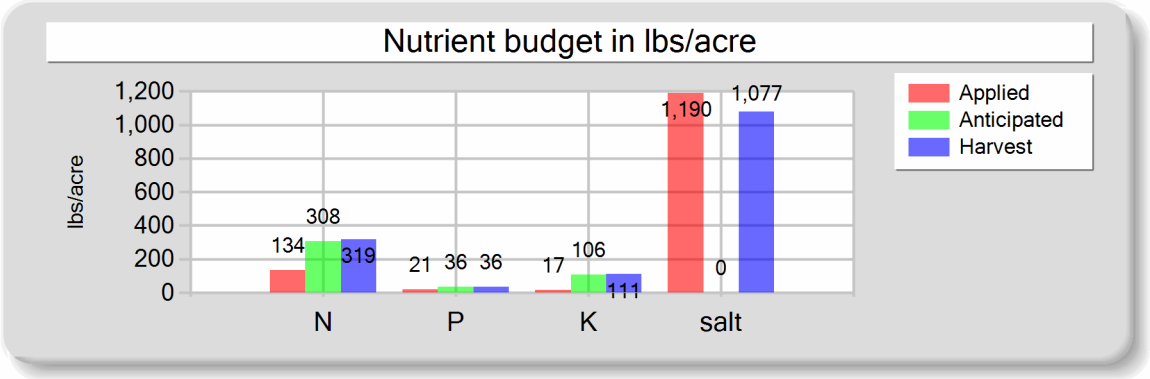
Fresh water applied	
17,339,500.00	*gallons*
638.55	*acre-inches*
33.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

G12 - 10/30/2022: Winter Forage

Field name: G12 Crop: Winter Forage Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.34	20.88	16.78	1,189.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	134.34	20.88	16.78	1,189.88
Anticipated crop nutrient removal	308.00	36.40	106.40	0.00
Actual crop nutrient removal	318.52	36.36	110.78	1,076.78
Nutrient balance	-184.19	-15.49	-94.00	113.09
Applied to removed ratio	0.42	0.57	0.15	1.11

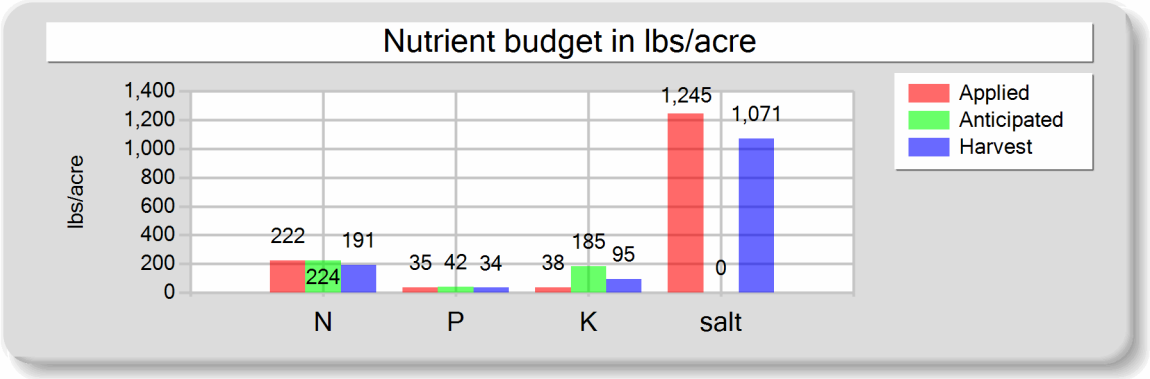
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G12 - 06/18/2023: Corn, silage

Field name: G12 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	79.00	10.00	10.00	0.00
Dry manure	136.37	24.71	27.90	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,244.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.37	34.71	37.90	1,244.87
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	191.36	34.23	95.15	1,070.92
Nutrient balance	31.01	0.48	-57.25	173.95
Applied to removed ratio	1.16	1.01	0.40	1.16

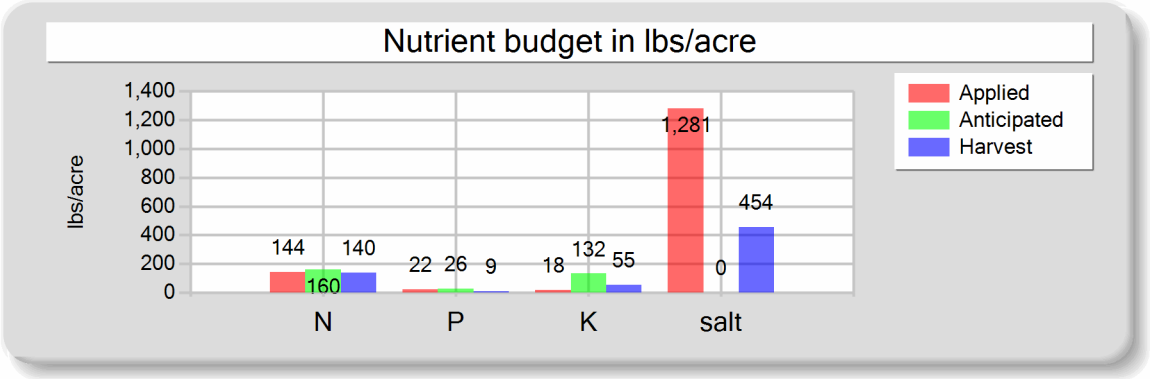
Fresh water applied
7,956,000.00 *gallons*
292.99 *acre-inches*
36.62 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G13 - 10/31/2022: Oats, hay

Field name: G13 Crop: Oats, hay Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.13	22.48	18.07	1,281.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.13	22.48	18.07	1,281.40
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	140.10	8.80	55.26	454.08
Nutrient balance	4.04	13.68	-37.19	827.32
Applied to removed ratio	1.03	2.55	0.33	2.82

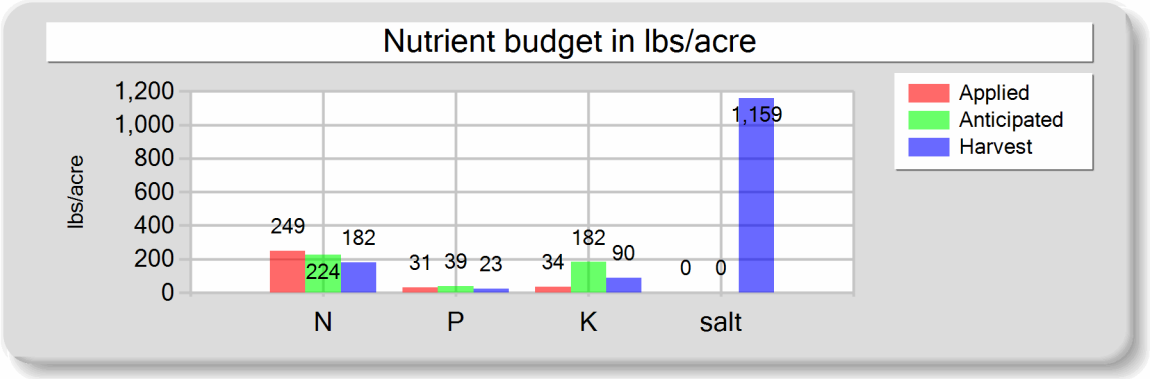
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G13 - 06/17/2023: Corn, silage

Field name: G13 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	79.00	10.00	10.00	0.00
Dry manure	117.49	21.29	24.04	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	45.39	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.88	31.29	34.04	0.00
Anticipated crop nutrient removal	224.00	39.20	182.00	0.00
Actual crop nutrient removal	181.73	23.37	89.74	1,159.15
Nutrient balance	67.15	7.92	-55.70	-1,159.15
Applied to removed ratio	1.37	1.34	0.38	0.00

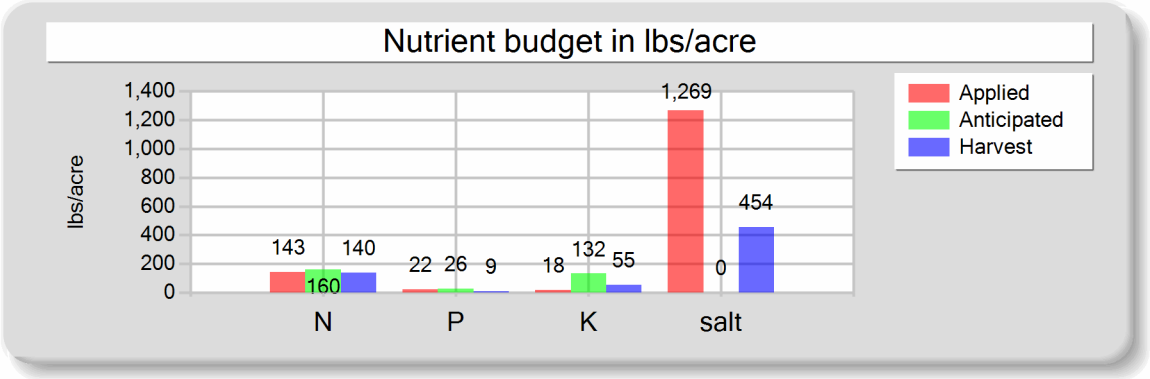
Fresh water applied	
20,145,000.00	*gallons*
741.87	*acre-inches*
28.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

G14 - 10/31/2022: Oats, hay

Field name: G14 Crop: Oats, hay Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	135.83	22.27	17.90	1,269.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	142.83	22.27	17.90	1,269.20
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	140.10	8.80	55.26	454.08
Nutrient balance	2.73	13.47	-37.36	815.12
Applied to removed ratio	1.02	2.53	0.32	2.80

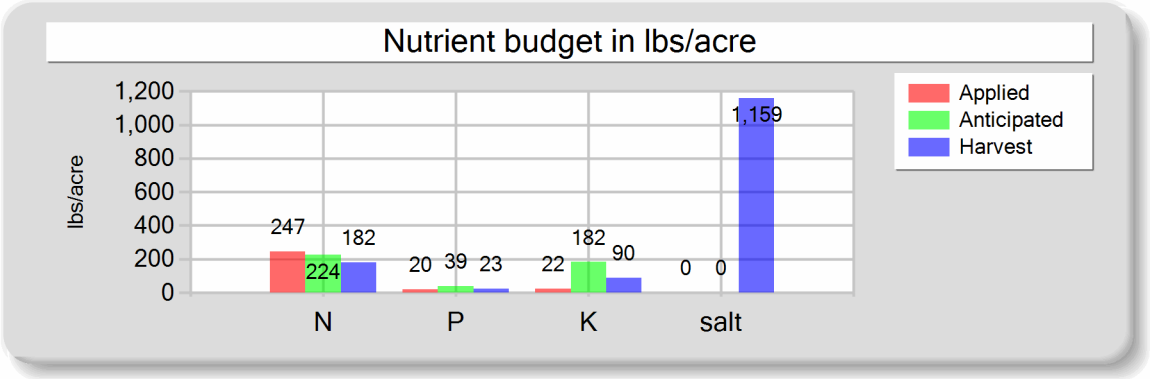
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

G14 - 06/11/2023: Corn, silage

Field name: G14 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	109.10	19.77	22.32	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	55.57	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.67	19.77	22.32	0.00
Anticipated crop nutrient removal	224.00	39.20	182.00	0.00
Actual crop nutrient removal	181.73	23.37	89.74	1,159.15
Nutrient balance	64.94	-3.60	-67.42	-1,159.15
Applied to removed ratio	1.36	0.85	0.25	0.00

Fresh water applied	
14,229,000.00	*gallons*
524.01	*acre-inches*
34.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2350014D

Sample and source description: T2350014D

Sample date: 12/16/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,500.00	5,000.00	4,020.00	18,100.00	1,950.00	990.00	4,800.00	1,090.00		28.50
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132029B

Sample and source description: T3132029B

Sample date: 05/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	4,730.00	5,340.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T2313012A

Sample and source description: T2313012A

Sample date: 11/09/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	434.00	319.00	0.00	0.00	41.30	543.00								6,100.00	4,480
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3059010A

Sample and source description: T3059010A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	302.00	0.00	0.00	69.80	553.00								5,410.00	6,700
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132029A

Sample and source description: T3132029A

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	515.00	344.00	0.00	0.00	35.40	553.00								6,370.00	4,860
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Dairy/Dom.

T3251017A

Sample description: T3251017A

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.00	0.00	9.00								1,080.00	0
DL	1.00	0.50	1.00								100.00	1

G30 Irr. Well

G30 Irr. Well

T3235023B

Sample description: T3235023B

Sample date: 08/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.03	0.00	1.03								349.00	0
DL	0.50	0.50	0.50								100.00	1

G8 Irr. Well

T3235023C

Sample description: T3235023C

Sample date: 08/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	7.10	4.54	43.20	114.00	0.00	11.60	12.40	280.00	150
DL	0.14	0.50	0.14	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

North Irr. Well

T3235023A

Sample description: T3235023A

Sample date: 08/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.32	0.00	0.32								286.00	0
DL	0.10	0.50	0.10								100.00	1

South Irr. Well

South Irr. Well

T3335010A

Sample description: T3335010A

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.02	0.00	7.02								739.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

G01

T3354016A

Sample and source description: T3354016A

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00		0.00	0.00	
DL			1.00		1.00	1.00	

G02

T3354016B

Sample and source description: T3354016B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.00		0.00	0.00	
DL			1.00		1.00	1.00	

G03

G03

T3354016B

Sample and source description: T3354016B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.00		0.00	0.00	
DL			1.00		1.00	1.00	

G05

T3354016A

Sample and source description: T3354016A

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

G01 - 10/29/2022: Winter Forage

T3158058A

Sample and source description: T3158058A

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	2,580.00	7,860.00		7.64
DL	1.00	1.00	1.00		1.00

G01 - 06/12/2023: Corn, silage

G01 - 06/12/2023: Corn, silage

T3291030A

Sample and source description: T3291030A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,140.00	10,700.00		6.30
DL	1.00	1.00	1.00		1.00

G02 - 10/28/2022: Ryegrass, silage

T3146020A

Sample and source description: T3146020A

Sample date: 05/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	3,680.00	28,400.00		11.40
DL	1.00	1.00	1.00		1.00

G02 - 06/16/2023: Corn, silage

T3291030A

Sample and source description: T3291030A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,140.00	10,700.00		6.30
DL	1.00	1.00	1.00		1.00

G03 - 10/28/2022: Ryegrass, silage

T3146020A

Sample and source description: T3146020A

Sample date: 05/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	3,680.00	28,400.00		11.40
DL	1.00	1.00	1.00		1.00

G03 - 06/19/2023: Corn, silage

T3291030B

Sample and source description: T3291030B

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,350.00	12,800.00		6.20
DL	1.00	1.00	1.00		1.00

G05 - 10/27/2022: Ryegrass, silage

W3108007A

Sample and source description: W3108007A

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	5,200.00	44,000.00		13.10
DL	1.00	1.00	1.00		1.00

G06 - 10/27/2022: Ryegrass, silage

W3108007A

Sample and source description: W3108007A

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	5,200.00	44,000.00		13.10
DL	1.00	1.00	1.00		1.00

G06 - 06/24/2023: Corn, silage

T3291030B

Sample and source description: T3291030B

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,350.00	12,800.00		6.20
DL	1.00	1.00	1.00		1.00

G09 - 06/19/2023: Corn, silage

T3291030D

Sample and source description: T3291030D

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,950.00	5,420.00		6.10
DL	1.00	1.00	1.00		1.00

G11 - 10/29/2022: Winter Forage

T3158058A

Sample and source description: T3158058A

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	2,580.00	7,860.00		7.64
DL	1.00	1.00	1.00		1.00

G11 - 06/20/2023: Corn, silage

T3291030D

Sample and source description: T3291030D

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,950.00	5,420.00		6.10
DL	1.00	1.00	1.00		1.00

G12 - 10/30/2022: Winter Forage

T3158058A

Sample and source description: T3158058A

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	2,580.00	7,860.00		7.64
DL	1.00	1.00	1.00		1.00

G12 - 06/18/2023: Corn, silage

T3291030D

Sample and source description: T3291030D

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,950.00	5,420.00		6.10
DL	1.00	1.00	1.00		1.00

G13 - 10/31/2022: Oats, hay

T3158058B

Sample and source description: T3158058B

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	1,250.00	7,850.00		6.45
DL	1.00	1.00	1.00		1.00

G13 - 06/17/2023: Corn, silage

T3291030C

Sample and source description: T3291030C

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,720.00	1,250.00	4,800.00		6.20
DL	1.00	1.00	1.00		1.00

G14 - 10/31/2022: Oats, hay

T3158058B

Sample and source description: T3158058B

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	1,250.00	7,850.00		6.45
DL	1.00	1.00	1.00		1.00

G14 - 06/11/2023: Corn, silage

T3291030C

Sample and source description: T3291030C

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,720.00	1,250.00	4,800.00		6.20
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

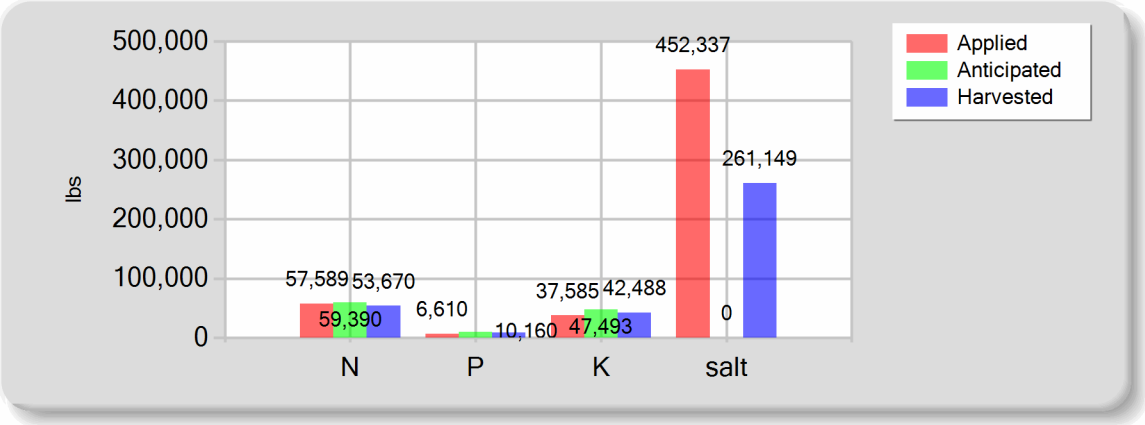
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

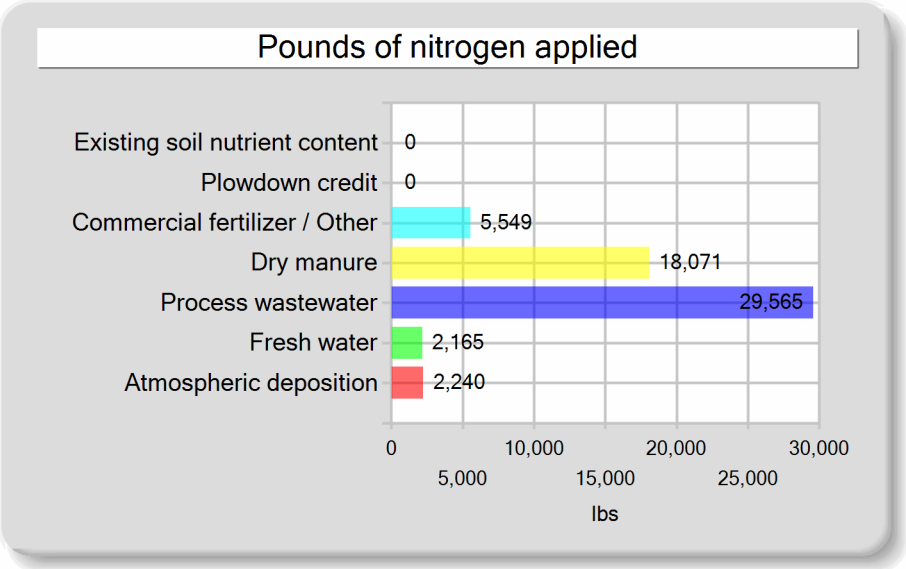
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

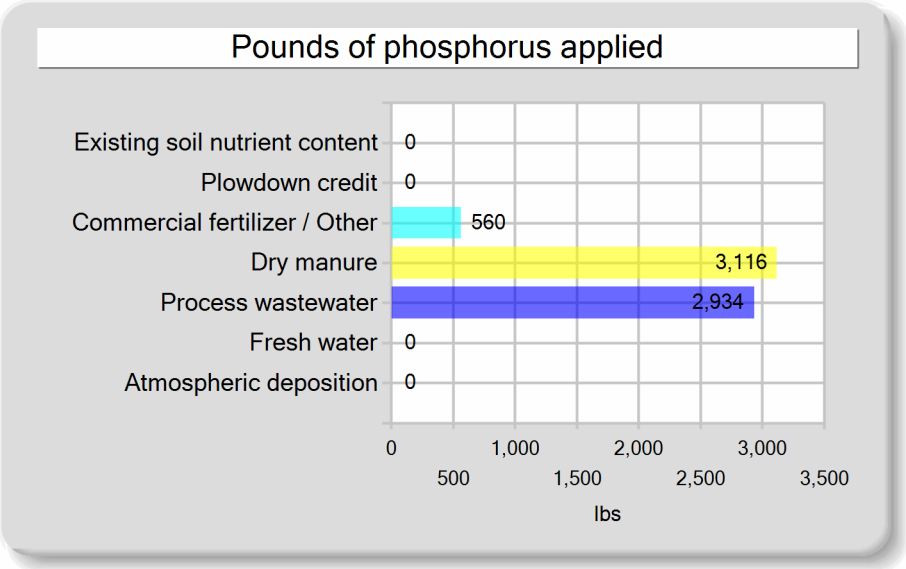
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,549.00	560.00	560.00	0.00
Dry manure	18,070.70	3,116.35	3,029.82	85,671.00
Process wastewater	29,564.58	2,933.86	33,994.90	330,852.55
Fresh water	2,164.66	0.00	0.00	35,813.19
Atmospheric deposition	2,240.00	0.00	0.00	0.00
Total nutrients applied	57,588.94	6,610.21	37,584.72	452,336.74
Anticipated crop nutrient removal	59,390.40	10,160.40	47,493.20	0.00
Actual crop nutrient removal	53,669.80	8,162.44	42,488.06	261,149.45
Nutrient balance	3,919.13	-1,552.23	-4,903.34	191,187.29
Applied to removed ratio	1.07	0.81	0.88	1.73

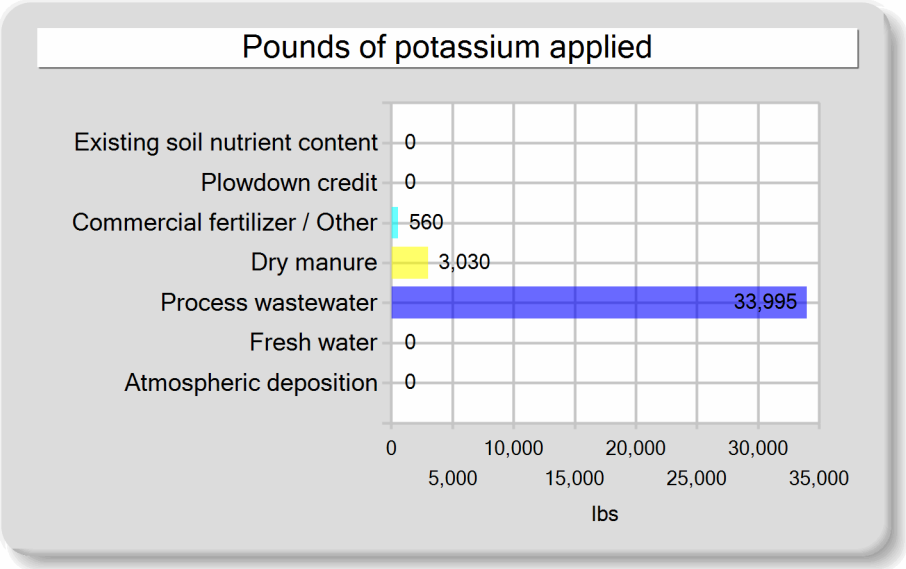
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

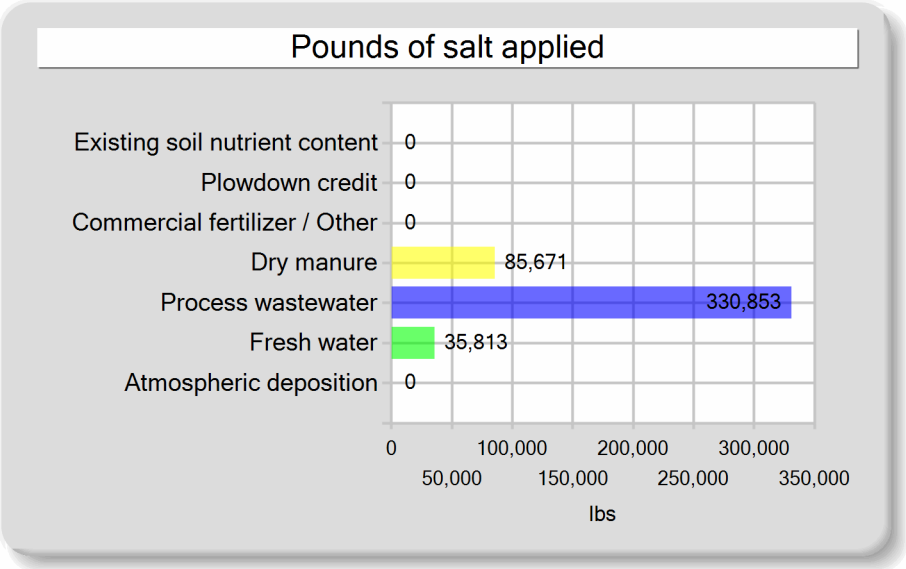


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

There were changes in administration of this facility during this year. Samples were inadvertently missed. 2024 will not have that issue.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Case A Van Steyn	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/14/24	6/14/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: CASE VAN STEYN

Name of Dairy Facility: Van Steyn Dairy #2

Facility Address:

Number and Street	City	County	Zip Code
13039 Pellandini RD	Galt	Sacramento	95632-8818

Contact Person Name and Phone Number: CASE VAN STEYN (Name) (209) 417-1010 (Phone Number)

MANURE HAULER INFORMATION

Name of Hauling Company/Person: CASE VAN STEYN

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
11823 CARROLL RD	ELK GROVE	CA	95757

Contact Person: CASE VAN STEYN (Name) (209) 417-1010 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

VAN STEYN DAIRY #1 (Name) (209) 417-1010 (Phone Number)

Address	City	State	Zip Code
11823 CARROLL RD	ELK GROVE	CA	95757

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
11823 CARROLL RD	ELK GROVE	95757

Street and nearest cross street (if no address) — County: Sacramento

X132-X332-X055-XXXX (Assessor's Parcel Number) — Sacramento (Assessor's Parcel Number County)

Last date hauled: 10/31/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 775.00 tons

Manure Solids Content: 20.9 %

Method used to determine amount of manure:

HAULER'S ESTIMATION

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator Signature — Date: 4-15-24

Hauler Signature — Date: 4-15-24

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: VAN STEYN DAIRY #2
Mailing address: 13039 PELLANDINI RD.
(not facility address) GALT, CA 95632
Phone#: (209) 745-1428

Authorized Copy Released To:
JACQUIE CAPTEIN RECORD KEEPING, INC. (JCRK, Inc.)
(209) 327-0992
Sampled By: Georgina Van Steyn
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH3-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH3-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/17/23	1:00p	North Irr Well	Ag Well	22.0°		X				
2	✓	1:30p	G30 Irr Well	✓	21.0°		X				
3	✓	2:00p	G8 Irr Well	✓	21.0°						H14
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		8/18/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	8-23-23 1:00			T3235023

(LABORATORY USE ONLY)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Van Steyn Dairy #2
13039 Pellandini Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 22.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
North Irr. Well	T3235023A	Electrical Conductivity (EC)	EPA 120.1 and	0.286	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.321	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	183	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	95.8 (T3234039B)	95.7 (T3234039B)	0.125
Electrical Conductivity (EC)	98.0			0.286
Nitrate Nitrogen (NO3-N)	104			7.24

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Van Steyn Dairy #2
13039 Pellandini Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
G 30 Irr. Well	T3235023B	Electrical Conductivity (EC)	EPA 120.1 and	0.349	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.03	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	224	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	95.8 (T3234039B)	95.7 (T3234039B)	0.125
Electrical Conductivity (EC)	98.0			0.286
Nitrate Nitrogen (NO3-N)	104			7.24

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Compliance Analysis Report

Customer
Van Steyn Dairy #2
13039 Pellandini Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 8/24/23

Irrigation /Ground Water (H14)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
G8 Irr. Well	T3235023C	Electrical Conductivity (EC)	EPA 120.1 and	0.280	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	114	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	11.6	mg/L
		Sodium (Na)	EPA 200.7	43.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	4.54	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	150	mg/L
		Calcium (Ca)	EPA 200.7	7.10	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	12.4	mg/L
		Soluble Salts (SALT-SOL)	Calculation	179	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.6	95.8 (T3234039B)	95.7 (T3234039B)			0.125
Calcium (Ca)	98.6			71.7 (T3235029A)	70.8 (DUP)	1.25
Chloride (Cl)	97.5	87.6 (W3236004A)	91.4 (W3236004A)			4.20
Magnesium (Mg)	98.8			37.0 (T3235029A)	36.5 (DUP)	1.33
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)			2.36
Sodium (Na)	100			87.4 (T3235029A)	88.7 (DUP)	1.48
Sulfate (SO4)	102	119 (W3236004A)	111 (W3236004A)			6.79
Total Dissolved Solids (TDS)	100			215 (T3235024A)	225 (Dup)	4.55

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DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: VAN STEYN DAIRY #2
Mailing address: 13039 PELLANDINI RD.
(not facility address) GALT, CA 95632
Phone#: (209) 745-1428

Authorized Copy Released To:
JACQUIE CAPTEIN RECORD KEEPING, INC.
(209) 327-0992
Sampled By: Georgina Van Steyn
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH3-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH3-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/7/23	4:00p	Dairy/Dom.	Dairy/Dom	21.4°	15 min	X				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jacquie Capteon	JCRK		9/8/23 8:00a
2nd		DENELE ANALYTICAL, INC.	9-8-23 1:00	

(LABORATORY USE ONLY)

Logged In By:

LABORATORY NUMBER: T3251017

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Van Steyn Dairy #2
13039 Pellandini Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/8/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom.	T3251017A	Electrical Conductivity (EC)	EPA 120.1 and	1.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.00	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	689	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	100	95.6 (W3248007A)	92.9 (W3248007A)	2.88

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Van Steyn Dairy #2
Mailing address: 13039 Pellandini Rd.
(not facility address) Galt, CA 95632
Phone#: (209) 745-1428

Authorized Copy Released To: JCRK, INC.
(209) 327-0992
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	11/30/23	2:00p	South Irr. Well	Irr. Well	14.6	20 min	x				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		12/1/23 8:00a		
2nd	Lana Gaston	DENELE ANALYTICAL, INC.	12-01-23 14:45		LG	T3335010

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Van Steyn Dairy #2
13039 Pellandini Road
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/1/23 3:02 PM
SUBMITTED BY: DENELE COURIER
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temp was above qualifying at 14.6C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
South Irr. Well	T3335010A	Electrical Conductivity (EC)	EPA 120.1 and	0.739	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.02	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	473	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	95.8			0.040
Nitrate Nitrogen (NO3-N)	95.3	109 (T3335007C)	105 (T3335007C)	3.39

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ELAP Certificate No.: 2714